



ICMC 2024-2025

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- 1 Joe the Jaguar is on an infinite grid of unit squares, starting at the centre of one of them. At the k -th minute, Joe must jump a distance of k units in any direction. For which n is it possible for Joe to be inside or on the edge of the starting square after n minutes?
- 2 Alice and the Mad Hatter are playing a game. At the start of the game, three 2024's are written on the blackboard. Then, Alice and the Mad Hatter alternate turns, with the Mad Hatter starting. On the Mad Hatter's turn, he must pick one of the numbers on the blackboard and increase it by 1. On Alice's turn, she must:
 - pick one of the numbers on the blackboard and decrease it by 1, and then
 - replace the two numbers a and b on the blackboard which were not chosen by the Mad Hatter on the previous turn with $\sqrt{ab}$.
 Alice wins if, on the start of her turn, any of the three numbers are less than 1. Can the Mad Hatter prevent Alice from winning?
- 3 Let V be a subspace of the vector space $\mathbb{R}^{2 \times 2}$ of 2-by-2 real matrices. We call V nice if for any linearly independent $A, B \in V$, $AB \neq BA$. Find the maximum dimension of a nice subspace of $\mathbb{R}^{2 \times 2}$.
- 4 Let a chain denote a row of positive integers which continue infinitely in both directions, such that for each number n , the n numbers directly to the left of n yield n distinct remainders upon division by n .
 - (a) If a chain has a maximum integer, what are the possible values of that integer?
 - (b) Does there exist a chain which does not have a maximum integer?
- 5 A positive integer is a non-trivial perfect power if it can be expressed as n^k where n and k are positive integers and $k > 1$. Show that there exist arbitrarily large consecutive square numbers with no other non-trivial perfect powers between them.
- 6 A set of points in the plane is called rigid if each point is equidistant from the three (or more) points nearest to it.
 - (a) Does there exist a rigid set of 9 points?
 - (b) Does there exist a rigid set of 11 points?

